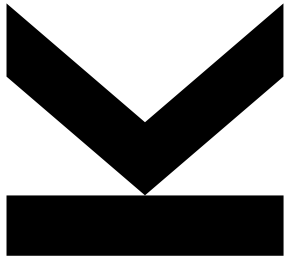
RTL Models in SystemC Labs



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V1.04

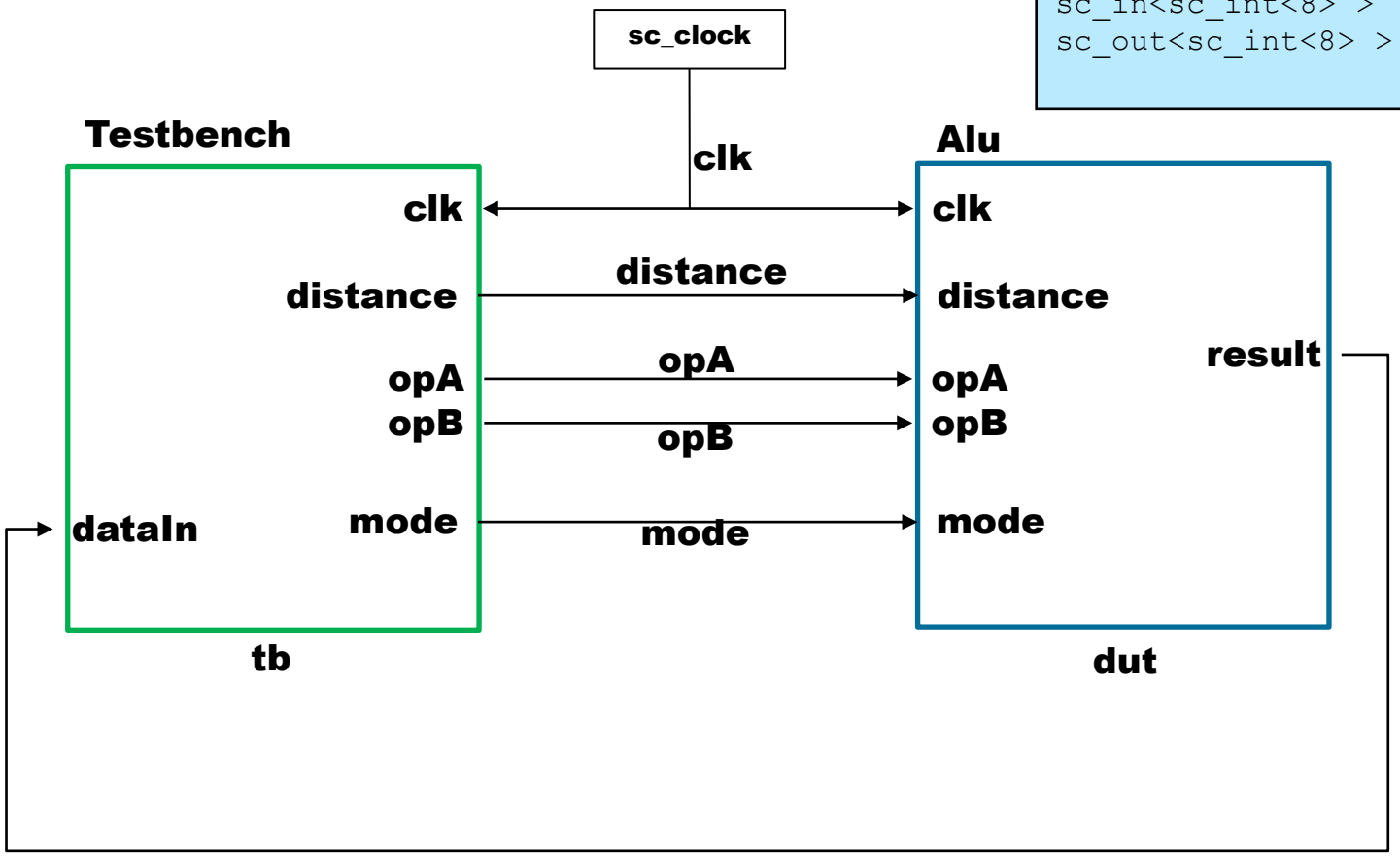
General

- Zip archive on Moodle: 3.2labs.zip
- Unpack to /home/user/ModernSystemDesign/
- Two directories in /home/user/ModernSystemDesign/3.2labs/
  - alu/
  - counter/
- Build: Open terminal, change to respective directory and

```
mkdir build
cd build
cmake ..
make
```

ALU

```
sc_in<bool>          clk;
sc_in<sc_lv<2> >     mode;
sc_in<sc_uint<4> >   distance;
sc_in<sc_int<8> >    opA, opB;
sc_out<sc_int<8> >   result;
```



ALU Specification

Result depends on „mode“:

00: Sum of `opA` and `opB`

01: Shift left `opA` for `distance` (opA)

10: Rotate `opA` left 1 bit

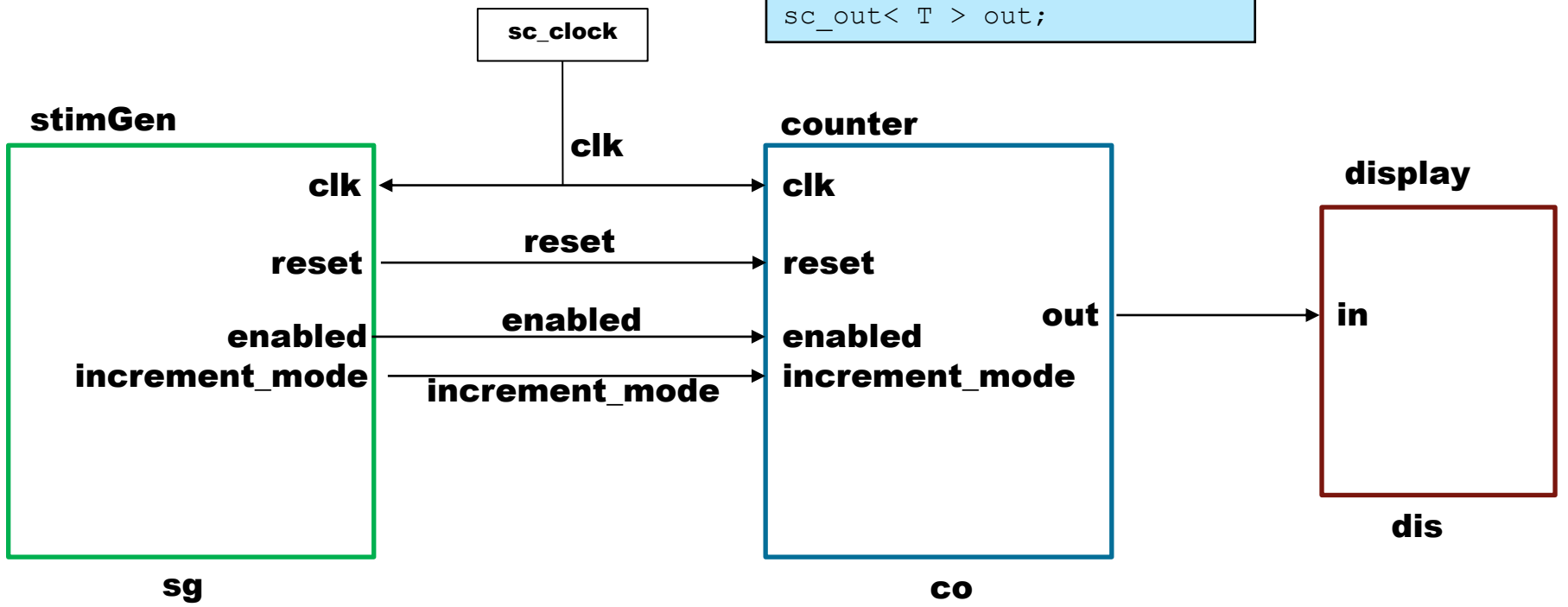
11: Subtract `opB` from `opA`

Assignment

- Complete the ALU code following the specification
- Reference output: expected.txt
- Think about likely cornercases (Overflows, …)

Counter Architecture

```
sc_in_clk clock;
sc_in<bool> reset;
sc_in<bool> enabled;
sc_in<bool> increment_mode;
sc_out< T > out;
```



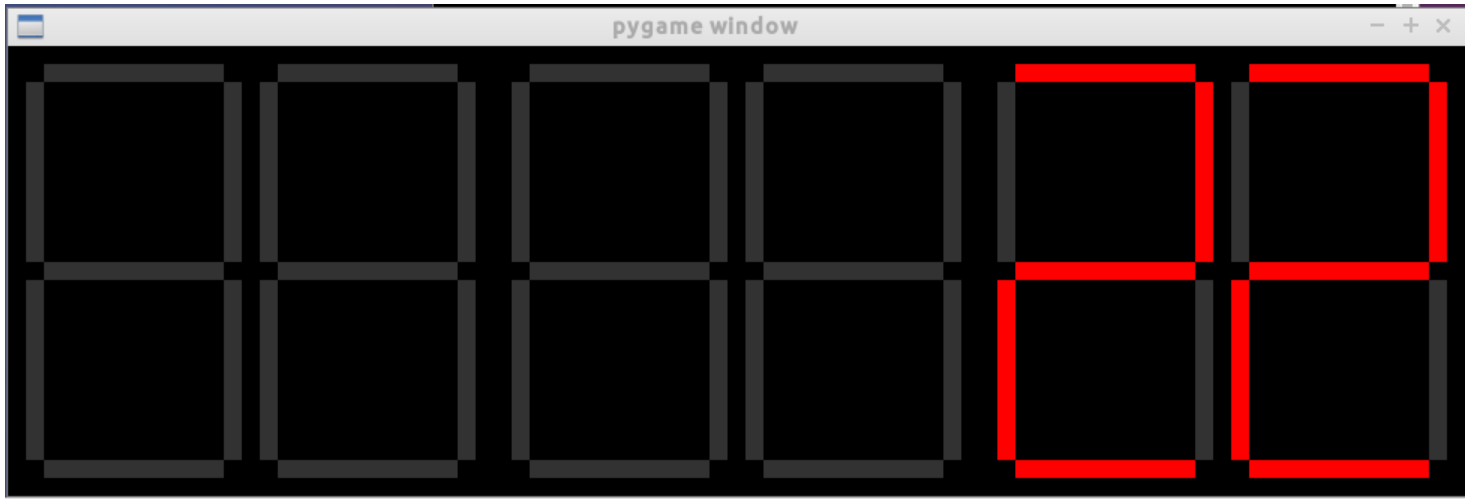
Assignment 2: 7-Segment Display Counter

- Inside „counter“ sub folder is the implementation of a 7-segment counter module

- Assignment:
  - Build a clock using the counter module
    - (The counter module must be modified for this)
  - Relevant questions:
    - Output
    - Simulation
    - Stimuli

Execute

- Counter waits for TCP connection from Python client on port 30000
  - Start Python client by running „`./client.py`“ AFTER starting „`counter`“



Execute (2)

- First, the counter starts counting up

- Then, StimGen starts polling the keyboard
  - Increment counter (w)
  - Decrement counter (s)